INFOSYS SE QUANT VAULT • RATIOS, PROPORTIONS, AVERAGES & AGES

Weighted Averages, Age Differences, Compounded Ratios, Alligation • Total Solved Questions: 22

[Campus Monk Day 01 Mock Test]

Topic: Quantitative Aptitude

Q1. 720 litres of a mixture contain milk and water only in the ratio of x : 3 respectively. If 30 litres water is added into it, then the ratio of quantity of milk to that of water in the resultant mixture will become 3:2. Find the value of

(A) 10

Correct Answer: See Logic

■ **Step-by-Step Logic:** Step 1: Set up the ratio equations based on the problem statement. Step 2: Calculate the initial quantities using the given proportion. Step 3: Account for the addition/removal of components and equate to the resultant ratio. Step 4: Solving the linear equation yields the target value, confirming .

[Campus Monk Day 01 Mock Test]

Topic: Quantitative Aptitude

Q2. If x is the mean of the first 8 positive integers multiple of 9 and Y is the Median for the same , then find the value of 5X-3Y.

(A) 81

(C) -XGog

Correct Answer: See Logic

■ **Step-by-Step Logic:** Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is .

[Campus Monk Day 01 Mock Test]

Topic: Quantitative Aptitude

Q3. What is the mean of the mode and median of the data given below? 8, 7, 5, 4, 9, 1, 3, 7, 6, 5, 2, 1, 6, 3, 9, 8, 4, 6

(A) 6.5

(B) 6.25

(C) 5.5

(D) 5.75

Correct Answer: Option (D) : 5.75

■ **Step-by-Step Logic:** Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (D) (5.75).

[Campus Monk Day 03 Mock Test]

Topic: Quantitative Aptitude

Q4. The ratio of the age of Ram and Rahim 10 years ago was 1 : 3. The ratio of their age five years hence will be 2 : 3. Then the ratio of their present age is

(A) 1 : 2

(B) ---

(C) 3 : 4

(D) 2 : 5

Correct Answer: Option (B) : ---

■ **Step-by-Step Logic:** Step 1: Set up the ratio equations based on the problem statement. Step 2: Calculate the initial quantities using the given proportion. Step 3: Account for the addition/removal of components and equate to the resultant ratio. Step 4: Solving the linear equation yields the target value, confirming Option (B) (---).

[Campus Monk Day 06 Mock Test]

Topic: Quantitative Aptitude

Q5. Seven years ago, the ratio of the ages of A and B was 2: 3. Three years hence, the ratio of their ages will be 5: 7. The sum of their present ages (in years) is:

(A) 100

(B) 200

(C) 112

(D) 114

Correct Answer: Option (D) : 114

■ **Step-by-Step Logic:** Step 1: Set up the ratio equations based on the problem statement. Step 2: Calculate the initial quantities using the given proportion. Step 3: Account for the addition/removal of components and equate to the resultant ratio. Step 4: Solving the linear equation yields the target value, confirming Option (D) (114).

[Campus Monk Day 07 Mock Test]

Topic: Quantitative Aptitude

Q6. Two containers of milk contain mixtures of water and milk in ratio 5:4 and 7:9. In what ratio they should be mixed so that mixture is of 6:6 ratios?

(A) A. 6:5

(B) A. 7:5

(C) A. 8:9

(D) A. 9:8

Correct Answer: Option (D) : A. 9:8

■ **Step-by-Step Logic:** Step 1: Set up the ratio equations based on the problem statement. Step 2: Calculate the initial quantities using the given proportion. Step 3: Account for the addition/removal of components and equate to the resultant ratio. Step 4: Solving the linear equation yields the target value, confirming Option (D) (A. 9:8).

[Campus Monk Day 08 Mock Test]

Topic: Quantitative Aptitude

Q7. Question : If the surface areas of two spheres are in the ratio of 49:81 , then find the ratio of their volumes.

(A) 27:64

(B) 125:512

(C) 343:729

(D) 216:512

Correct Answer: Option (C) : 343:729

■ **Step-by-Step Logic:** Step 1: Set up the ratio equations based on the problem statement. Step 2: Calculate the initial quantities using the given proportion. Step 3: Account for the addition/removal of components and equate to the resultant ratio. Step 4: Solving the linear equation yields the target value, confirming Option (C) (343:729).

[Campus Monk Day 08 Mock Test]

Topic: Quantitative Aptitude

Q8. Question : A mixture contains milk and water in ratio 5 : 7, when 8 liters milk of and 8 liters of water is added in mixture then ratio becomes 9 : 11. Find initial quantity of milk.

(A) 24 liters

(B) 10 liters

(C) 16liters

(D) 18 liters

Correct Answer: Option (B) : 10 liters

■ **Step-by-Step Logic:** Step 1: Set up the ratio equations based on the problem statement. Step 2: Calculate the initial quantities using the given proportion. Step 3: Account for the addition/removal of components and equate to the resultant ratio. Step 4: Solving the linear equation yields the target value, confirming Option (B) (10 liters).

[Campus Monk Day 08 Mock Test]

Topic: Quantitative Aptitude

Q9. Question : The mean weight of 34 students of a school is 42 kg. If the weight of the teacher is also included, the mean rises by 400 grams. Find the weight of the teacher (in kg.)

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is .

[Campus Monk Day 08 Mock Test]

Topic: *Quantitative Aptitude*

Q10. Rearrange them to form a logically coherent paragraph 1■■ Start-ups are increasingly adopting artificial intelligence to automate routine operations and reduce human error. 2■■ Investors are keen to fund ventures that can demonstrate scalable AI-driven business models. 3■■ As competition intensifies, companies seek innovative tools to gain a market edge. 4■■ However, reliance on AI also raises ethical concerns about transparency and job displacement. Q1. After rearranging into a coherent paragraph, which sentence will come second? (Enter the integer only)

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Step 1: Set up the ratio equations based on the problem statement. Step 2: Calculate the initial quantities using the given proportion. Step 3: Account for the addition/removal of components and equate to the resultant ratio. Step 4: Solving the linear equation yields the target value, confirming .

[Campus Monk Day 08 Mock Test]

Topic: *Quantitative Aptitude*

Q11. Rearrange them to form a logically coherent paragraph 1■■ Start-ups are increasingly adopting artificial intelligence to automate routine operations and reduce human error. 2■■ Investors are keen to fund ventures that can demonstrate scalable AI-driven business models. 3■■ As competition intensifies, companies seek innovative tools to gain a market edge. 4■■ However, reliance on AI also raises ethical concerns about transparency and job displacement. Q2. Which sentence best serves as the concluding idea for the paragraph? (Enter the integer only)

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Step 1: Set up the ratio equations based on the problem statement. Step 2: Calculate the initial quantities using the given proportion. Step 3: Account for the addition/removal of components and equate to the resultant ratio. Step 4: Solving the linear equation yields the target value, confirming .

[Campus Monk Day 08 Mock Test]

Topic: *Quantitative Aptitude*

Q12. Each sentence below has a blank space. Four options are suggested. Select the most appropriate word/phrase to complete the sentence meaningfully. Mark the correct option. Despite several challenges, the new metro project has made remarkable progress and is now expected to be completed _____ schedule.

(A) beneath

(B) ahead of

(C) on

(D) over

Correct Answer: Option (B) : ahead of

■ **Step-by-Step Logic:** Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (B) (ahead of).

[Campus Monk Day 08 Mock Test]

Topic: *Quantitative Aptitude*

Q13. Choose the word that is nearest in meaning to the underlined word. Question The detective's report was meticulous , leaving no room for doubt. careless detailed hasty approximate

(A) careless

(B) detailed

(C) hasty

(D) approximate

Correct Answer: Option (B) : detailed

■ **Step-by-Step Logic:** Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (B) (detailed).

[Campus Monk Day 09 Mock Test]

Topic: *Quantitative Aptitude*

Q14. The ratio of the age of Ram and Rahim 10 years ago was 1 : 3. The ratio of their age five years hence will be 2 : 3. Then the ratio of their present age is

(A) 1 : 2

(B) > ■

(C) 3 : 4

(D) 2 : 5

Correct Answer: Option (B) : > ■

■ **Step-by-Step Logic:** Step 1: Set up the ratio equations based on the problem statement. Step 2: Calculate the initial quantities using the given proportion. Step 3: Account for the addition/removal of components and equate to the resultant ratio. Step 4: Solving the linear equation yields the target value, confirming Option (B) (> ■).

[Campus Monk Day 09 Mock Test]

Topic: *Quantitative Aptitude*

Q15. How many 3-letter words with or without meaning, can be formed out of the letters of the word, 'ADVERTISING', if repetition of letters is not allowed?

(A) Explanation: The word 'ADVERTISING' has 10 different letters. Hence, the number of 3-letter words(with or without meaning) formed by using these letters = 10P3

(B) 420

(C) 5040

(D) None of these

Correct Answer: Option (A) : Explanation: The word 'ADVERTISING' has 10 different letters. Hence, the number of 3-letter words(with or without meaning) formed by using these letters = 10P3

■ **Step-by-Step Logic:** Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (A) (Explanation: The word 'ADVERTISING' has 10 different letters. Hence, the number of 3-letter words(with or without meaning) formed by using these letters = 10P3).

[Campus Monk Day 09 Mock Test]

Topic: *Quantitative Aptitude*

Q16. In each of the questions given below four words are given in bold. These words may or may not be in their correct position. Following each sentence four sequences are provided. Select the sequence of the words which will make the given sentence contextually meaningful. If the words are correct at their current position, then choose option (e) as your answer. India has an advantage (A), as a large origin (B) of semiconductor design engineers globally (C) are either Indian or Indian- portion (D).

(A) DCBA

(B) BADC

(C) ADCB

(D) CADB

Correct Answer: Option (C) : ADCB

■ **Step-by-Step Logic:** Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (C) (ADCB).

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q17. In each of the questions given below four words are given in bold. These words may or may not be in their correct position. Following each sentence four sequences are provided. Select the sequence of the words which will make the given sentence contextually meaningful. If the words are correct at their current position, then choose option (e) as your answer. Pollution has badly native (A) the fish harvest and has destroyed (B) the breeding grounds (C) of the affected (D) fish.

(A) ABDC
(B) CBAD
(C) DBCA
(D) BDAC

Correct Answer: Option (C) : DBCA

■ **Step-by-Step Logic:** Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (C) (DBCA).

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q18. Directions : In each of the question given below a/an idiom/phrase is given in bold which is then followed by five options which then try to decipher its meaning. Choose the option which gives the correct meaning of the phrases Larger than life

(A) to be cruel
(B) to be haughty
(C) to be impressive and important
(D) to be disrespectful

Correct Answer: Option (C) : to be impressive and important

■ **Step-by-Step Logic:** Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (C) (to be impressive and important).

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q19. Directions : out of the four alternatives, choose the one which best expresses the meaning of the given word. Our posterity shall hold us responsible for the devastation of Ozone layer.

(A) predecessors
(B) ---
(C) offsprings
(D) followers

Correct Answer: Option (B) : ---

■ **Step-by-Step Logic:** Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (B) (---).

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q20. Directions : ,out of the four alternatives, choose the one which best expresses the meaning of the given word. The ambition of being an IAS is subdued by the poverty within.

(A) modified
(B) challenged
(C) affected
(D) ---

Correct Answer: Option (D) : ---

■ **Step-by-Step Logic:** Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (D) (---).

[College ITM Mock Paper 5]

Topic: Mathematical Critical Thinking

Q21. A container has 80 litres of pure milk. 20 litres of the mixture is removed and replaced with water; this process is repeated once more. How much pure milk remains?

(A) 45 litres
(B) 50 litres
(C) 40 litres
(D) 48 litres

Correct Answer: Option (A) : 45 litres

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 6]

Topic: Mathematical Critical Thinking

Q22. In what ratio must milk-water mixtures of strength 3:2 and 5:1 (milk:water) be mixed to obtain a mixture of strength 2:1?

(A) 5:2
(B) 2:5
(C) 3:2
(D) 1:3

Correct Answer: See Logic

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.